Agenda

- Syllabus Update
- Have you started HW1?
- Coroutines
- Channels
- Coroutine implementation

Clarification

The exact behavior of yield (does it **have** to switch?) is defined by the specific coroutine library

HW1: Yes

Code examples today: No (more general)

Channels

- A crucial part of coroutines
- How any useful work gets done!

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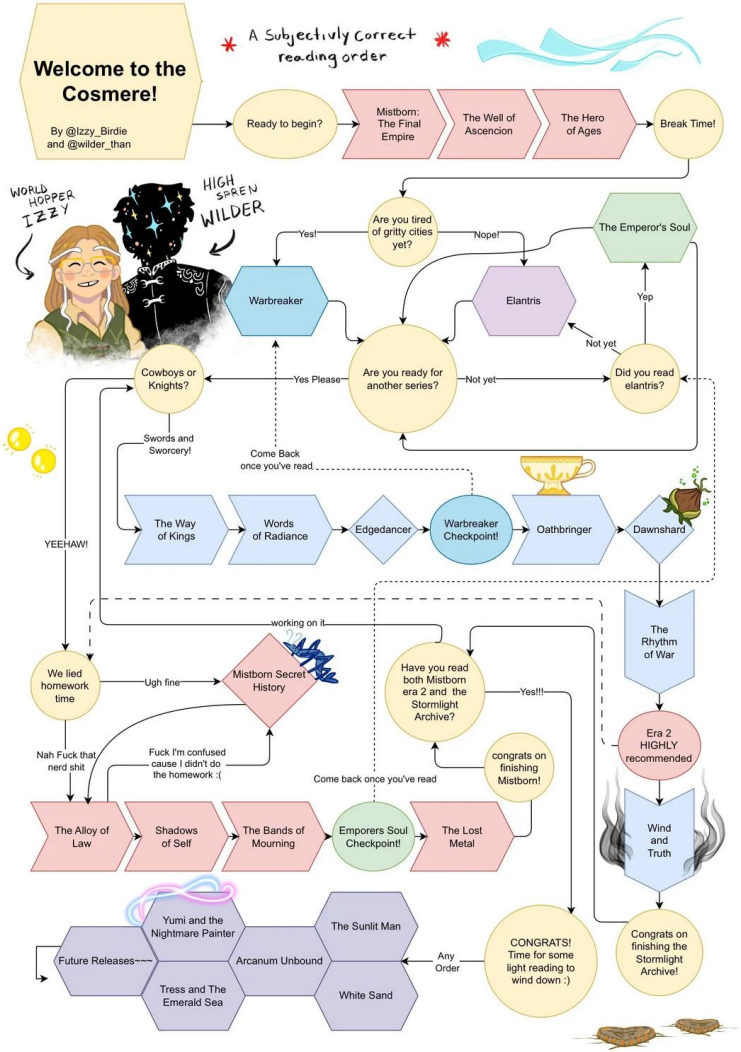

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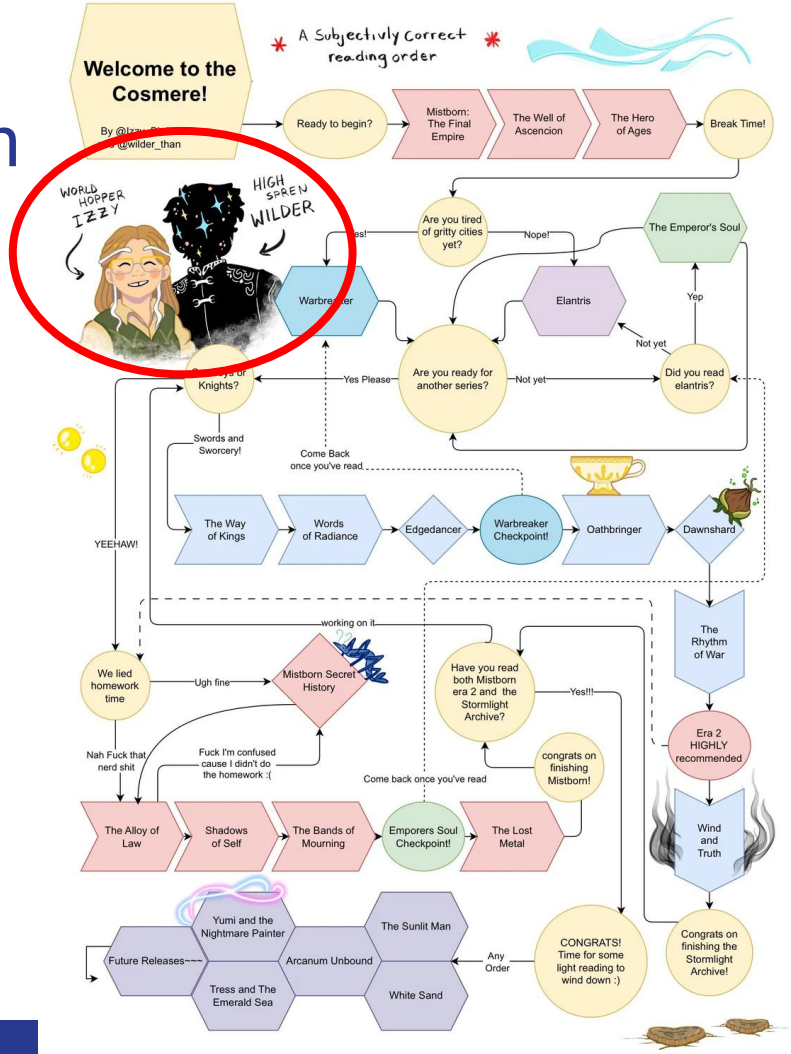

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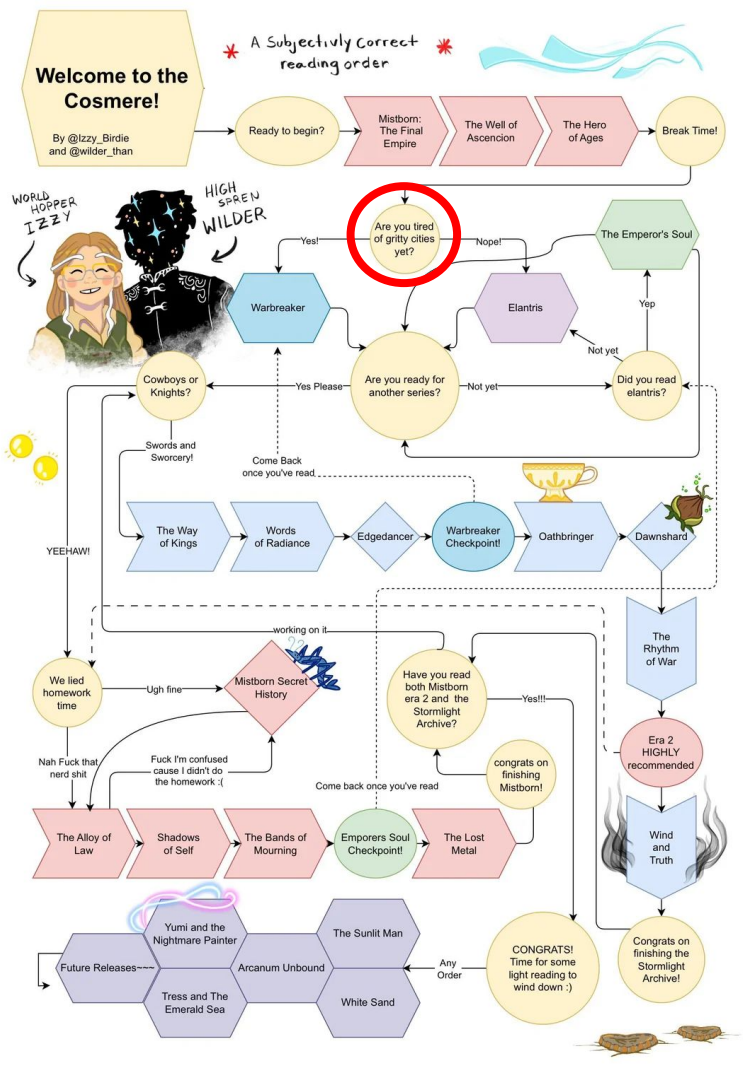

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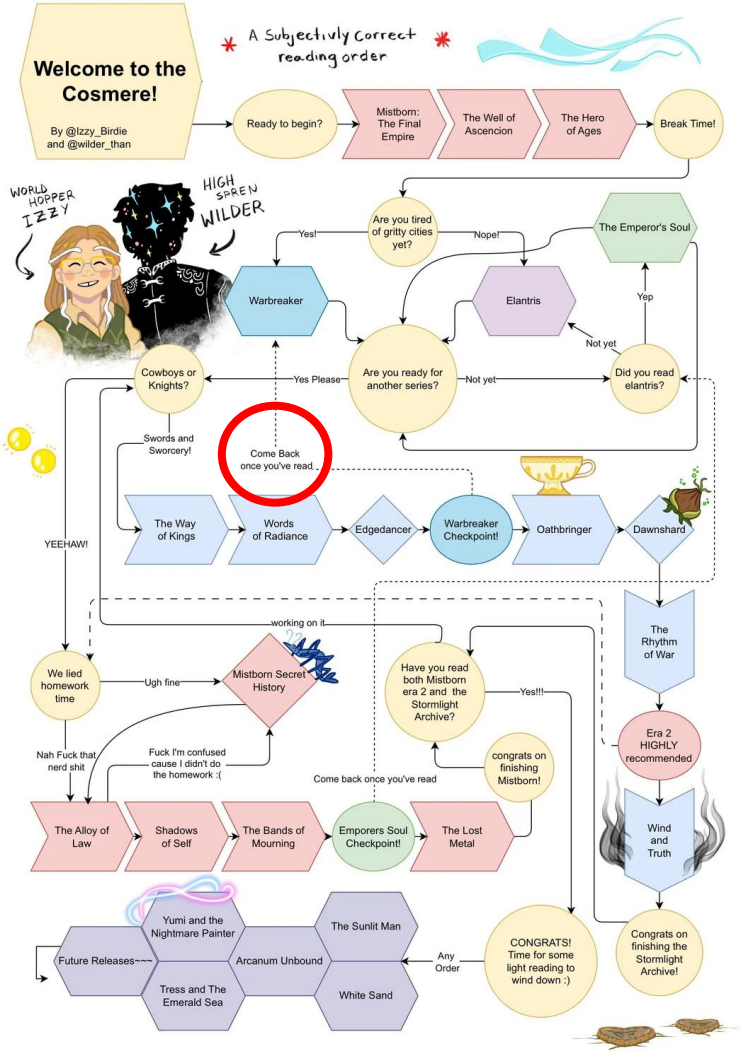

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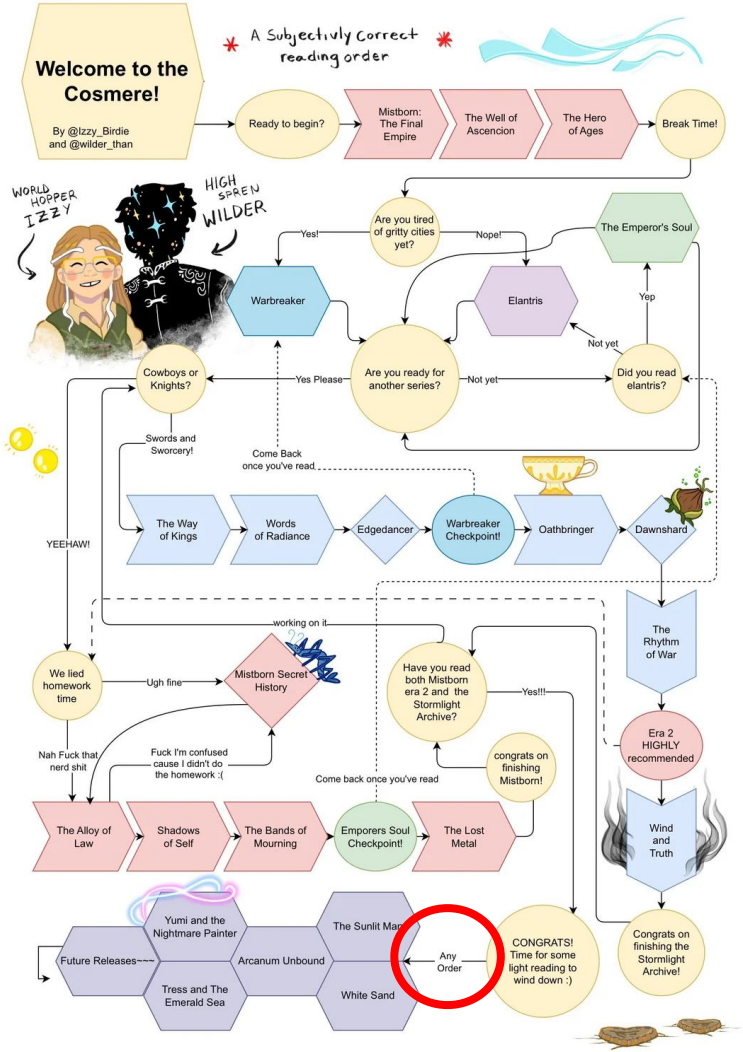

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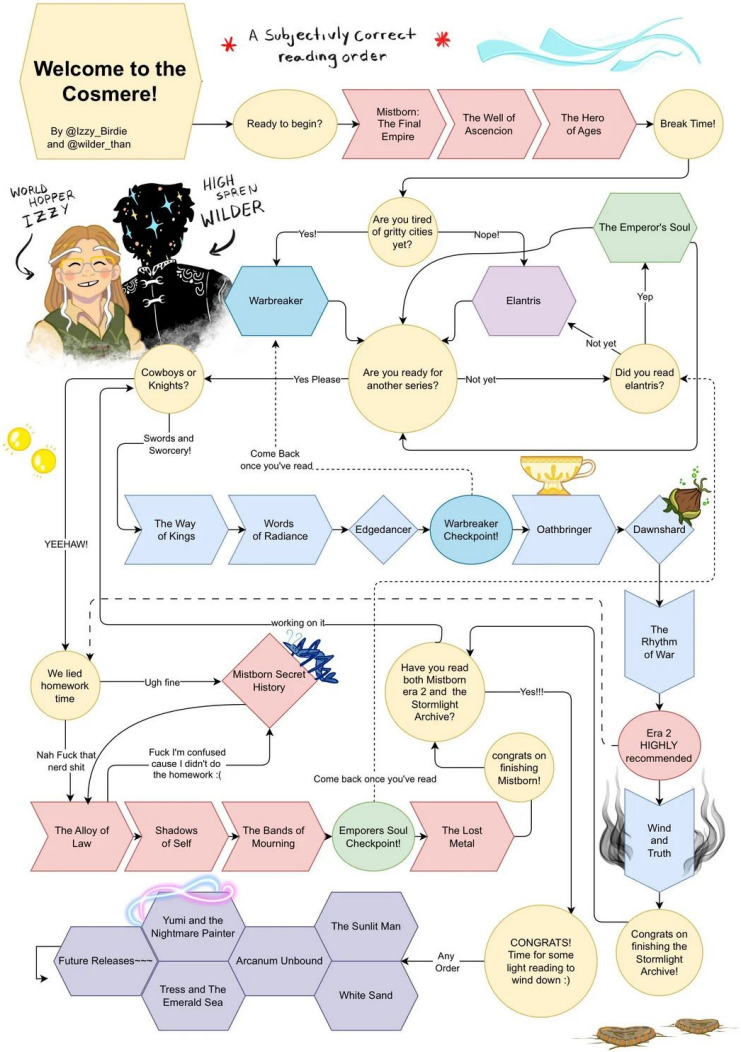
What does this remind you of?



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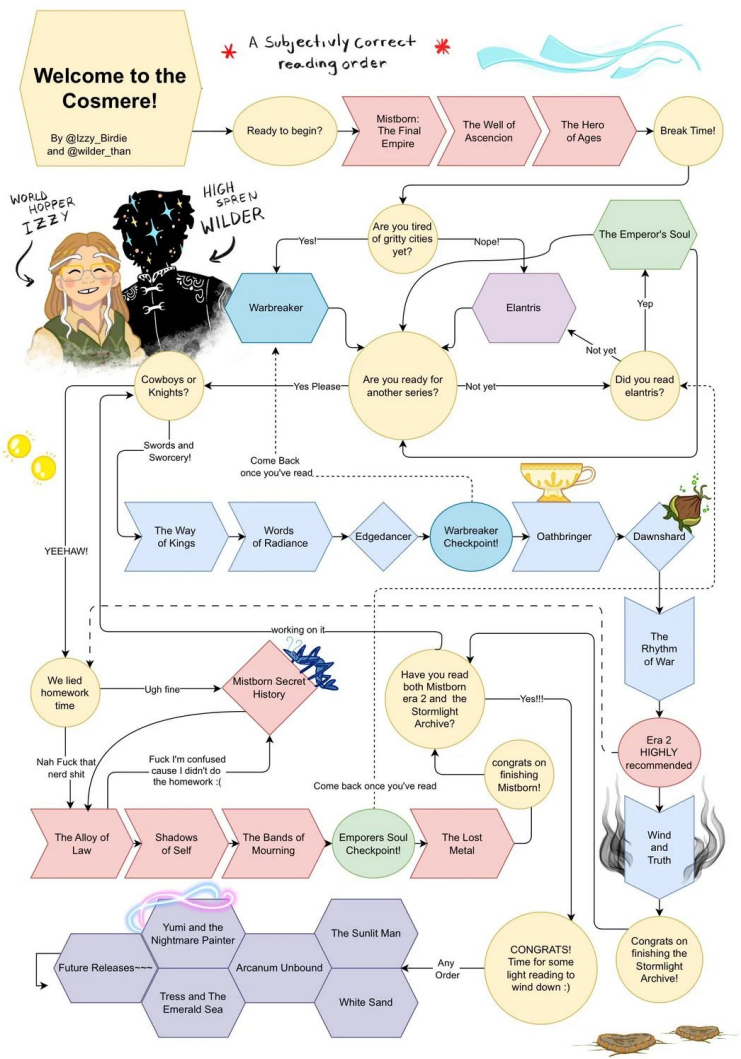
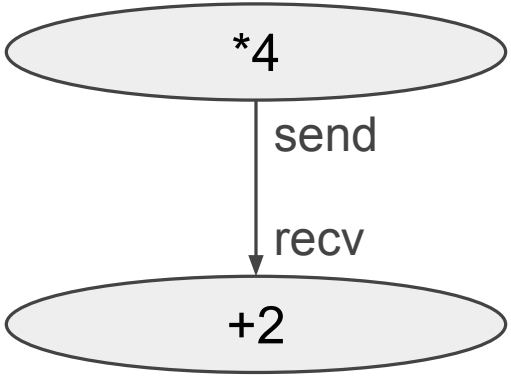
What does this remind you of?

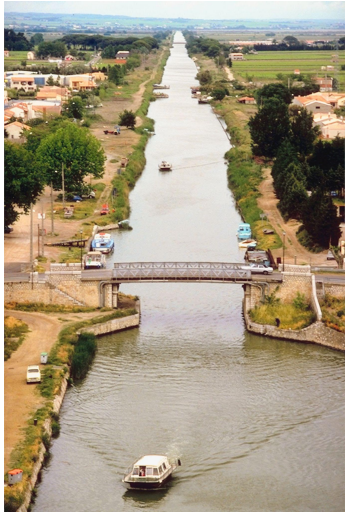
Dependency Graphs!



Channels

- How you implement dependencies
- Synchronize data transfer between coroutines!
  - send and recv





Example 3

```
fn main() {
    h, c = spawn(my_routine)
    send(c, 5)
    join(h)
}
```

```
fn my_routine(c) {
    x = recv(c)
    print(x)
}
```

Example 3

```
fn main() {
    PC
    h, c = spawn(my_routine)

    send(c, 5)

    join(h)

}
```

```
fn my_routine(c) {

    x = recv(c)

    print(x)

}
```

Example 3

```
fn main() {

    h, c = spawn(my_routine)

    send(c, 5)

    join(h)

}
```

```
fn my_routine(c) {
    PC
    x = recv(c)

    print(x)

}
```

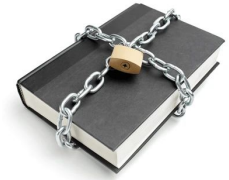
Example 3

```
fn main() {

    h, c = spawn(my_routine)
    PC
    send(c, 5)

    join(h)

}
```



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fn my_routine(c) {

    x = recv(c)

    print(x)

}
```

Example 3

```
fn main() {

    h, c = spawn(my_routine)

    send(c, 5)
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    join(h)

}
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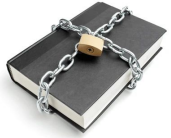
Example 3



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fn main() {

    h, c = spawn(my_routine)

    send(c, 5)

    join(h)

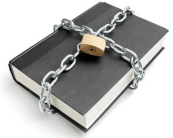
}
```

```
fn my_routine(c) {

    x = recv(c)
    PC
    print(x)

}
```

Example 3



```
fn main() {

    h, c = spawn(my_routine)

    send(c, 5)

    join(h)

}
```

```
fn my_routine(c) {

    x = recv(c)

    print(x)
```

PC

```
}
```

Example 3

```
fn main() {

    h, c = spawn(my_routine)

    send(c, 5)

    join(h)
```

PC

```
}
```

```
fn my_routine(c) {

    x = recv(c)

    print(x)

}
```

Example 4

```
fn main() {
    h, c = spawn(numbers)
    for i in 0..5 {
        x = recv(c)
        print(x)
    }

}
```

```
fn numbers(c) {
    x = 0; y = 1;
    loop {
        sum = x + y
        send(c, sum)
        x = y
        y = sum
    }
}
```

Many flavors of channels

- Are send/recv buffered?
  - Can you send into a buffer (to be matched with a receiver later), or do you need a matching receiver immediately to finish the transaction?
    - HW1 is the latter (non-buffered channels)
- What type of data do they send?
  - In HW1, it will be a (void*), aka a pointer to anything
- What type of data is safe to send?

Communication without channels is unsafe

```
my_global = 1

fn main() {

    h, c = spawn(my_routine)

    print(my_global)

    join(h)

}
```

```
fn my_routine(c) {

    my_global = 2

}
```

Channels create causal consistency

If a program proceeds past a send/recv, then we know something about where 2 coroutines are

Channels create causal consistency

```
my_global = 1

fn main() {

    h, c = spawn(my_routine)

    _ = recv(c)

    print(my_global)

    join(h)

}
```

```
fn my_routine(c) {

    my_global = 2

    send(c, null)

    ...

}
```

Channels create causal consistency

Say we want to create 100 coroutines, but we want **only** main() to run until we are ready to “release” all of the coroutines. How do we do this?

- Hint: you can make arbitrary channels via create_channel()
- Hint: you can do this in O(1) space!

Channels create causal consistency

```
fn main() {
    big_c = create_channel()
    for i in 0..100 {
        h, c = spawn(my_routine)
        send(c, big_c)
    }
    ... once ready:
    for i in 0..100 {
        send(big_c, 1);
    }
}
```

```
fn my_routine(c) {
    big_c = recv(c)
    _ = recv(big_c)
    ... do work here
}
```

Why use Coroutines?

Coroutines sometimes could run in parallel (e.g. Go) (but not homework 1)

Coroutines allow you to program things in a more natural way

- Each function can take care of a specific task, but execution can return to main without the function needing to fully finish

Coroutines are cheap

- Can run hundreds of thousands of coroutines simultaneously

Coroutines enable asynchronous behavior

Why use Coroutines?

“Parallelize Waiting”

While waiting for a disk I/O or network API calls, you can do something else

Relationship to Async/Await

- Async/Await abstraction builds on top of coroutines
  - Automatically places yields, sometimes hides sending and receiving too

Go

One of the most popular emerging programming languages

Their gimmick is coroutines, which they playfully call “goroutines”

Go

`f(x)` to run f

`go f(x)` to run f in its own coroutine

Go

```go
func main() {
    ch := make(chan int)
    go worker(ch)

    r := <-ch
}
```

```go
func worker(ch chan int) {
    fmt.Print("working...")
    time.Sleep(time.Second)
    fmt.Println("done")

    ch <- 42
}
```

And now…

You’re basically a Go expert!*

But the principle of Coroutines applies in many languages

Now…

We will talk about the implementation of a Coroutine in C. It will be highly relevant to Homework 1.